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Masar Health + Thaniya

Right Patient. Right Care. Right Place. Right Time.

FlowCare / MED-GWPYB3

Healthcare Crowd & Patient Flow Management

Filter 2: Prototype • Demonstration with simulated data

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Problem & Solution

Problem

During mass gatherings, patients may choose care without knowing clinical urgency or facility capability. Sudden casualty surges add pressure on emergency teams and receiving hospitals.

Solution

Masar combines structured symptom screening with care routing. Thaniya adds a simulated casualty coordination workflow. Clinical urgency takes priority over congestion. The prototype supports decisions and does not diagnose.

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Evidence of Need

WHO identifies sudden demand beyond local medical resources as a defining challenge in mass casualty incidents. [1]

WHO guidance for mass gatherings emphasizes planning, emergency preparedness and coordinated response. [2]

Our design hypothesis: combining urgency, facility capability and capacity visibility may improve coordination.

Evidence gap: no local demand dataset, interviews or measured patient outcomes yet. We make no quantified impact claim.

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How It Works

1. Select a language, age group and gathering location.
2. Answer structured symptom and red-flag questions.
3. The server applies demo urgency rules and filters facility capability before considering travel time and simulated load.
4. Thaniya records dispatch, reserves capacity and confirms hospital handover.
5. The command center displays saved demo activity and simulated surge scenarios.

Emergency urgency takes priority over congestion. These rules require clinical validation before operational use.

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Innovation & Alternatives

Existing approaches
Routine and incident triage protocols already exist, including WHO IITT. [3] Maps help locate facilities, while coordination teams manage transport and receiving capacity.

Our proposed contribution
One demonstrator connects patient screening, capability-aware routing, incident status and command-center visibility. Low-acuity cases can be explored separately from emergency pathways.

Differentiation remains a hypothesis. We do not claim to be the first Saudi solution or to outperform deployed systems.

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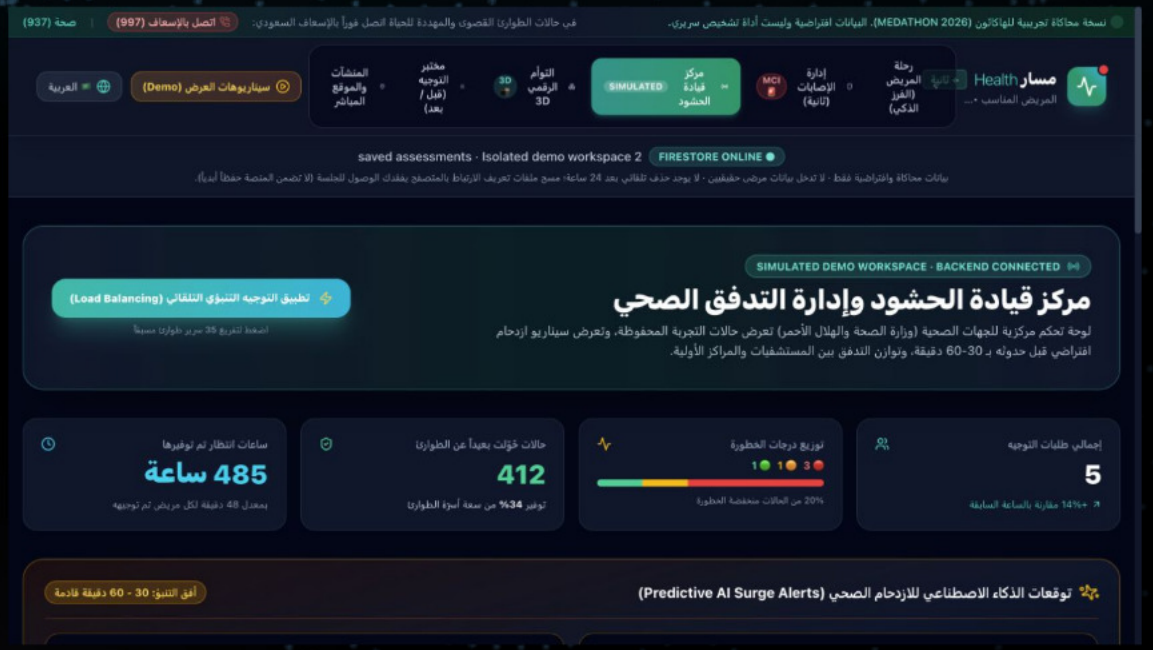
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Prototype / Proof of Concept

Working web prototype: patient screening, casualty coordination, command center, routing lab and a 3D demonstration.

The public frontend calls an Express backend. Tests cover saved assessments and casualty handover.

All patient cases, capacities and forecasts are simulated. No live hospital or ambulance integration.



https://masar-thaniya-flowcare.ai.studio/

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Data & Tools Used

Frontend: React and TypeScript with Vite; Firebase anonymous sign-in provides a browser-persistent demo identity.

Backend: Express verifies the Firebase ID token, applies demo triage and coordinates dispatch and handover. Firestore REST stores each owner's workspace with atomic version-checked updates.

Controls: owner-isolated rules, input validation, stale-write rejection and duplicate-handover protection. Owners can modify their own synthetic workspace; this is not clinical authentication.

Retention: no automatic 24-hour deletion. Clearing browser/site data can lose access. Permanent retention is not guaranteed.

Scope: simulated data only. No clinical validation or real hospital/ambulance integration. SQLite is an explicit local test option.

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Testing & Validation

Verified on 10 November 2026
Public API connectivity, high- and low-acuity demo paths, and saved state after browser reload. Hospital handover updated occupied and reserved beds together.

Automated technical checks
HTTP integration tests, TypeScript checks and production build passed. Tests cover isolated workspaces, stale writes, red flags and duplicate handover.

What this demonstrates
A working frontend and backend for synthetic scenarios.

Not yet established
Clinical accuracy, field usability, predictive performance, real integrations or improved patient outcomes.

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Feasibility & Risks

Requirements
A clinical lead, EMS and hospital workflow owners, secure hosting, verified capacity feeds and an agreed escalation process.

Main risks and proposed controls
Under-triage: clinician-approved rules, red-flag escalation and prospective safety tests.
Stale capacity: timestamps, expiry rules and manual confirmation.
Connectivity loss: clear fallback instructions and coordinator oversight.
Privacy and misuse: minimum data, access controls and audit logs before any real deployment.

The current prototype is for demonstration, not clinical operation.

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Adoption & Sustainability

Beneficiaries: pilgrims and visitors, field teams, hospital coordinators and event command centers.

Potential adopters: health clusters, event medical providers and emergency coordination organizations. No partnership or purchasing commitment is confirmed.

Adoption route: supervised simulation, usability review, then an authorized pilot with clinical governance.

Value to measure: routing time, capability mismatches, workload distribution and safety failures.

Sustainability proposal: an institutional subscription or event contract covering support, hosting and integrations. Pricing and willingness to pay remain untested.

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Business model

Key Partners

Potential EMS and hospital partners
Event operators
Clinical reviewers

Key Activities

Workflow design
Safety testing
Integration maintenance

Key Resources

Clinical oversight
Engineering
Verified facility data

Value Proposition

Urgency-aware routing
Shared incident status
Capacity visibility

Benefits require pilot evidence

Customer Relationships

Pilot onboarding
Training and support
Feedback reviews

Channels

Event partnerships
Institutional pilots
Direct procurement

Customer Segments

Health clusters
Event medical providers
Emergency coordinators

Patients use the service

Cost Structure

Hosting and API usage, engineering, clinical review, security, integration and training

Revenue Streams

Proposed annual institutional subscription or event license. No revenue or validated price yet.

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Next Steps

1. Clinical review — Ola
Recruit an emergency-care reviewer and define a synthetic test set with expected pathways.

2. Visitor and usability checks — Ghaida
Test Safari, Chrome and WhatsApp entry on independent devices. Record completion and failure points.

3. Incident simulation — Alanoud
Run a tabletop surge exercise with explicit capacity changes and handover events.

4. Engineering and pilot readiness — Zayed
Resolve defects, review privacy and arrange an authorized pilot. All milestones are proposed, subject to reviewer and partner availability.

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Team Members

FlowCare: proposed project responsibilities



Ola Mustafa Alhashim
Clinical review coordination



غيداء عمر الشريف
User research and usability



Alanoud Khalid Albanna
Incident workflow and testing



Zayed Mohameed alnefaie
Product and prototype delivery

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Sources

[1] WHO: Mass Casualty Management
https://www.who.int/teams/integrated-health-services/clinical-services-and-systems/emergency-and-critical-care/mass-casualty-management

[2] WHO: Managing health risks during mass gatherings
https://www.who.int/activities/managing-health-risks-during-mass-gatherings

[3] WHO: Interagency Integrated Triage Tool
https://www.who.int/tools/triage

References checked 10 November 2026. They support the problem context, not clinical validation of this prototype. Live demonstration and source code links appear on the final slide.

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Demo & Technical Review

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Live Demo
https://masar-thaniya-flowcare.ai.studio/

GitHub: Frontend + Backend source and tests
https://github.com/zayedalnefaie-spec/masar-thaniya-medathon